

**Practice** Midterm **Solutions**, Summer 2024

This test has **5** questions worth a total of **100** points.

**Read the submission instructions below carefully before starting the exam.**

1. First, read through the exam PDF carefully, and answer the questions you feel most comfortable about first.
2. The exam consists of five short-answer questions. You **must** submit your answers as a **typed PDF**, with **one question response per page**. That means that you should submit a five page PDF, with your answer to Question 1 on page 1, your answer to Question 2 on page 2, etc.
3. Just to make my life easier, at the top of each page you should write the question that you are answering on that page. For example, your first page should have **Question 1** written at the top.
4. The response to each of the five questions can be given as (at most) a short paragraph. Excessively long answers will be **penalized**.
5. If you fail to follow these submission guidelines (in addition to just making your exam harder to grade) I will **deduct** a few points from your final midterm score.
6. You **must** submit your PDF to this assignment on Canvas by 8:25PM (the end of class) on the day of the exam: any late exams will not be graded.
7. You **are** allowed to resubmit to the Canvas assignment as many times as you like if you find a mistake (**as long** as your final submission is before the end of the exam!)

## 1. Should This Work? [20 Points]

Identify whether the following piece of Python code should work or not. If it should work, give an explanation of why. If it should not work, explain why.

You **must** explain using concepts we have learned from the lectures (if you Google/ChatGPT and use terminology that we never learned...I will be able to tell and give you zero points).

Your answer should be **at least** a few sentences, but no longer than a paragraph. Essay answers or answers that are a few words (or one sentence) will receive no credit.

```
def mystery(funky):  
    for i in range(1000):  
        print("funky")  
        print("meow")  
  
    for i in range(100):  
        funky += 1  
        meow += 3  
  
oof = 100  
meow = 50  
mystery(oof)
```

This program does not work. The reason is because of variable scopes. When we are running code inside of a function (such as the `mystery` function here), we can see and modify local variables: these are the variables that would end up being stored in the function's "local frame". We can also see variables stored in the global frame, but not modify them. `meow` is one such variable here, and our attempts to modify it inside of the `mystery` function will cause this program to throw an error.

## 2. What Do I Need to Change? [20 pts]

The code below is a broken version of the solution to Lab 4, Question 2. Why is this solution broken, and what would we need to change in order to fix things? There may be more than one issue with the program: you should identify and fix all of them.

You **must** explain using concepts we have learned from the lectures (if you Google/ChatGPT and use terminology that we never learned...I will be able to tell and give you zero points).

Your answer should be **at least** a few sentences, but no longer than a paragraph. Essay answers or answers that are a few words (or one sentence) will receive no credit.

```
# TODO: Get the number of books purchased from the user.
number = int(input("How many books did you buy: "))

# TODO: Determine how many points were earned.
if number >= 0:
    points = 0
elif number >= 2:
    points = 5
elif number >= 4:
    points = 15
elif number >= 6:
    points = 30
elif number >= 8:
    points = 60

print('This earns you this many points:')
# TODO: Print the number of points earned (just the number).
print(points)
```

This program is broken because an if/elif chain will run the code for the first condition that evaluates to true. In this case, even an extremely high number of books bought will trigger the first case in the if (`number >= 0`). Because an if/elif chain runs the code for the first true condition, we actually need to reverse the order of the conditions here to fix things.

### 3. What Do I Need to Change? [20 pts]

The program below is intended to read in a number from the user, add 100 to it, and print the result. Why will this program not work? What needs to be changed about this program to get it to work? There may be more than one issue with the program: you should identify and fix all of them.

You **must** explain using concepts we have learned from the lectures (if you Google/ChatGPT and use terminology that we never learned...I will be able to tell and give you zero points).

Your answer should be **at least** a few sentences, but no longer than a paragraph. Essay answers or answers that are a few words (or one sentence) will receive no credit.

```
number_from_user = input("Enter a number please: ")
result_number = number_from_user + 100
print(Result_number)
```

This program is broken for two reasons. Firstly, note that `number_from_user` is actually a string (this is, by default, what `input` returns). We then attempt to add a string to a number on Line 2, which causes an error. Fixing this would require us to use `int()` on line 1 around the `input()` call. However, there is actually a second issue here: we attempt to print the `Result_number` variable on Line 3. However, no variable with this name exists, because variable names in Python are case sensitive! We should have printed `result_number` instead.

#### 4. Functionally and Fundamentally Quite Fun [20 Points]

Identify whether the following piece of Python code should work or not. If it should work, give an explanation of why. If it should not work, explain why.

You **must** explain using concepts we have learned from the lectures (if you Google/ChatGPT and use terminology that we never learned...I will be able to tell and give you zero points).

Your answer should be **at least** a few sentences, but no longer than a paragraph. Essay answers or answers that are a few words (or one sentence) will receive no credit.

```
def fun_function(first, second, third):  
    print(first)  
    print(second)  
    print(third)  
  
    print(x)  
    print(y)  
    print(z)  
  
x = 100  
y = 1000  
z = x + y  
fun_function(x, y)
```

This program does not work. The issue here is very straightforward: the definition/signature of `fun_function` specifies that the function actually takes three arguments when called. However, we have only given the function two arguments on the final line of the program. We would need to supply a third (likely the programmer meant to supply the `z` variable from the Global Frame as well here).

## 5. Scrambled (Like an Egg)

Identify whether the following piece of Python code should work or not. If it should work, give an explanation of why. If it should not work, explain why.

You **must** explain using concepts we have learned from the lectures (if you Google/ChatGPT and use terminology that we never learned...I will be able to tell and give you zero points).

Your answer should be **at least** a few sentences, but no longer than a paragraph. Essay answers or answers that are a few words (or one sentence) will receive no credit.

```
j = 0

for i in range(10000):
    print(i)
    j += 1
    j *= 2

print(x)
x = j + 7
```

This program does not work. The issue here is again straightforward. Every time a variable is used (in any language), it must actually exist. Here we can see that we print the x variable before actually creating it. It is likely that the programmer accidentally flipped the order of the last two lines of the program.

**You have reached the end.**

**There is no more.**

**Go outside after this and take a study break. You deserve it :)**

